Telecom Customer Churn Analysis

Business Summary Report: problem solved, key answers and profit actions

100,000	49.6%	$58.68	$2.88M
customers analysed	churn rate (49,562 customers)	average monthly revenue per customer	monthly revenue of churned customers

1. The business problem

A telecom operator is losing customers and does not know **why they leave, which customers are most likely to leave next, and what it should do about it.** Every churned customer takes monthly revenue with them, and winning a replacement customer usually costs more than keeping an existing one. This project analyses 100,000 customers (100 data fields on usage, billing, handsets, call quality, customer care, tenure and demographics) to answer three questions:

- **Why** are customers leaving?
- **Who** is most at risk, so that retention spending goes where it pays back?
- **What** should the business do, and how do we know it will make money?

2. Executive summary

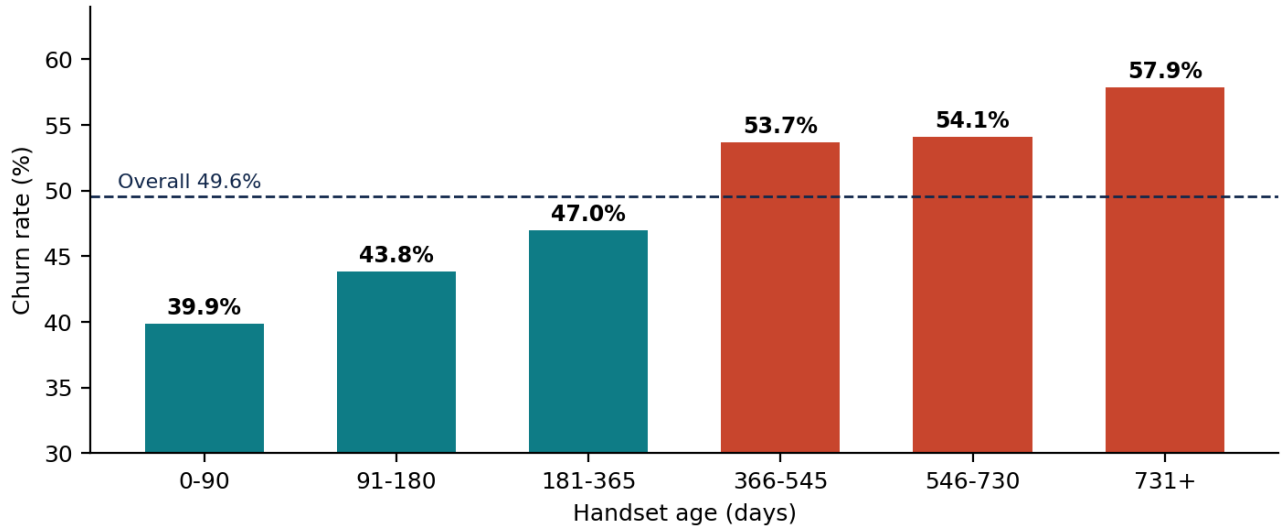
- **Handset age is the clearest warning sign.** Churn rises from 39.9% for handsets under 90 days old to 57.9% for handsets older than two years, with a visible jump after the one-year mark (47.0% to 53.7%).
- **Low-engagement customers leave more.** Churned customers use about 11% fewer minutes (477 vs 536 per month), and customers who never contact customer care churn at 51.8% against 44.2% for those who contact it six or more times.
- **Being a high spender does not protect a customer.** Churn is 52.6% in the lowest revenue quartile and 48.6% in the highest, so premium customers are still at risk and are worth more to save.
- **No single factor explains churn,** and demographics are only weakly linked to it. Retention works best by combining several signals into a risk score and acting on the highest-risk group.
- **Profit comes from precise targeting.** In the worked example in Section 6, a retention offer to 10,000 customers with handsets older than two years breaks even at a success rate of about 9.2%. Below that it loses money, above it the return grows quickly.

All findings describe associations in the data, not proven causes. Section 7 lists the limits.

3. Key business questions and answers

Business question	Answer from the data
Why are customers leaving?	Mostly handset age and type, falling usage and low engagement. No one factor dominates; the strongest single signal is handset age.
Does handset age matter?	Yes. Churn climbs steadily from 39.9% (0-90 days) to 57.9% (731+ days), with a jump past 365 days (47.0% to 53.7%).
Does the type of handset matter?	Yes. Handsets priced under $50 churn at 57.5% and over $200 at 36.0%. Web-capable-only (WC) devices churn at 56.3% vs 47.1% for WCMB devices, and customers with no recorded capability churn at 59.2%.
Does falling usage signal churn?	Modestly. Customers whose monthly minutes fall by more than 100 churn at 53.6%; those with growing usage churn at 46.2%.
Are high-revenue customers more loyal?	Only slightly. Lowest quartile 52.6% vs highest quartile 48.6%. Revenue alone does not make a customer safe.
Do customers who contact customer care churn more?	No, the opposite. 0 calls: 51.8%; 1-2 calls: 48.6%; 3-5 calls: 45.3%; 6+ calls: 44.2%. Silent customers look like the bigger risk. This is an association, and it overlaps with how heavily customers use the network.
When in the customer lifecycle is risk highest?	Newest customers (0-6 months) churn least at 38.4%. Risk peaks at 13-24 months (51.8%), around the time of typical upgrade and renewal decisions.
Does call quality matter?	A little. Churned customers complete fewer calls (68.0% vs 70.7%), but the gap is small, so network quality is a secondary factor.
Do demographics or region matter?	Only weakly. Every demographic association is small. Regional churn ranges from 45.9% (Midwest) to 56.9% (Northwest/Rocky Mountain).

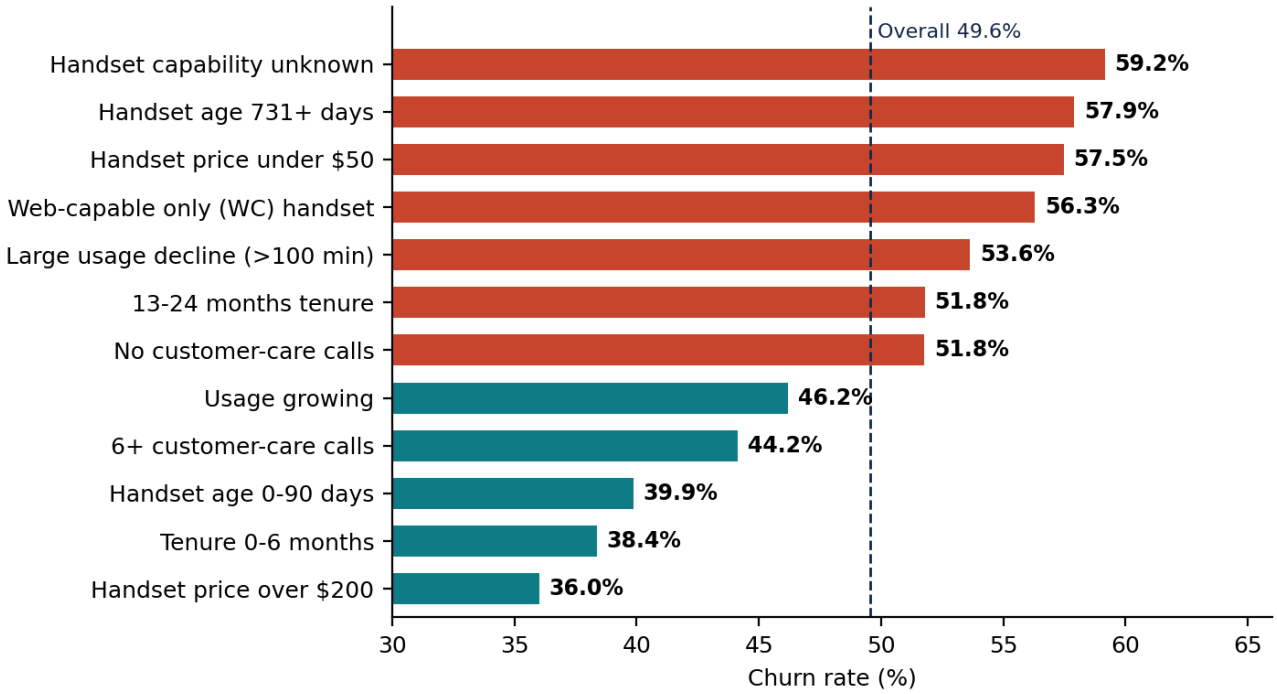
Churn rate by handset age



Teal bars are below the overall churn rate and red bars are above it.

4. Where the risk is concentrated

The chart compares the highest-risk and lowest-risk customer groups found in the analysis against the overall churn rate of 49.6%. The retention budget should be pointed at the red groups and the teal groups should be protected, not discounted.



How much revenue is at stake?

In this 100,000-customer sample, the 49,562 customers who churned paid an average of $58.16 a month, which is about **$2.88 million of monthly revenue (roughly $34.6 million a year)** that left with them. This dataset is a balanced sample with a much higher churn rate than a real carrier would see, so use the figure to size the problem and rescale it with your real churn rate before using it in a budget.

What this means in practice

- The customers most worth acting on are those with **old or basic handsets, falling usage and little contact with the company**.
- Customers who are new (0-6 months) or on newer, higher-priced handsets are the most loyal groups. Spending discounts on them wastes margin.
- The risk score from the model in the project notebook ranks customers by likelihood to churn, so offers go to the top of the list first. Model performance details are in the notebook.

5. Actionable steps to protect profit

Actions are listed in priority order. Each one targets a group identified in the data and is tied to a measurable result.

#	Action	Target group and evidence	How it protects profit	Measure
1	**Pre-expiry handset upgrade offer.** Contact customers before their handset reaches 12 months.	Handset age 366+ days churns at 53.7% to 57.9%, vs 39.9% for the newest.	A well-priced upgrade keeps the customer's monthly revenue. The subsidy must cost less than the margin kept.	Churn in the 366+ day group; offer take-up; margin per saved customer
2	**Move basic-handset users onto better devices** (trade-in, leasing or bundle).	Handsets under $50: 57.5% churn. WC devices: 56.3%. Capability unknown: 59.2%.	Customers on better handsets churn far less (36.0% above $200), so a better device lowers future churn.	Churn after upgrade; revenue per customer
3	**Usage-drop early warning.** Trigger an offer or call when monthly minutes fall by more than 100.	Large decline: 53.6% churn vs 46.2% for growth.	Acts before the customer decides to leave, so a smaller offer can work.	Win-back rate; offer cost per saved customer
4	**Proactive outreach to silent customers** who are also high-risk, using low-cost channels (SMS, app, email) before discounts.	0 care calls: 51.8% churn vs 44.2% for 6+ calls.	Contact is cheap compared with discounts and helps unhappy customers who would never call.	Response rate; churn of contacted vs not-contacted
5	**Renewal conversation at 12-24 months.** Protect the strong first six months with onboarding.	Tenure 13-24 months peaks at 51.8%; 0-6 months is lowest at 38.4%.	Keeps customers through the highest-risk window and builds long-term revenue.	Churn by tenure band; renewal rate
6	**Regional review** of the highest-churn area and test local offers.	Northwest/Rocky Mountain 56.9% vs Midwest 45.9%.	Local fixes (network, competitor offers) cost less than national discounts.	Churn in the test region vs control
7	**Capture missing customer data** at sign-up and in service.	Missing capability data goes with 59.2% churn.	Better data improves the risk score and lets offers be matched to customers.	Share of unknown fields

Rule for every action: test it on a small group with a control group first, spend only on customers with a high risk score, and keep spend per customer below the margin that customer is expected to earn.

6. Worked profit example (illustrative)

This example shows how to decide whether a retention offer will make money. Items marked "data" come from the analysis. Items marked "assumption" are placeholders and must be replaced with the company's real numbers.

Input	Value	Source
Customers targeted (handset age 731+ days)	10,000	Example size
Expected churn without action	57.9% (about 5,789 customers)	Data
Average monthly revenue per customer	$58.68	Data
Gross margin on revenue	40%	Assumption
Months a saved customer stays	12	Assumption
Offer cost per customer reached	$15 ($150,000 in total)	Assumption
Margin kept per saved customer per year	$281.66 (58.68 x 40% x 12)	Calculated

Result by offer success rate

Success rate (share of would-be churners kept)	Customers saved	Margin kept	Net result after $150,000 cost	Return on spend
5%	289	$81,528	-$68,472	-45.6%
10%	579	$163,055	$13,055	8.7%
15%	868	$244,583	$94,583	63.1%
20%	1,158	$326,111	$176,111	117.4%
25%	1,447	$407,638	$257,638	171.8%

Break-even success rate: about 9.2%. A blanket offer that saves fewer than about 1 in 11 would-be churners loses money. The offer pays off when it is aimed at high-risk customers, costs less per customer, or is combined with cheaper actions (Actions 3 and 4) first.

7. Limits and next steps

Limits

- The data is a balanced sample (about 50% churn). Real carriers see much lower churn, so recalibrate the risk score and rescale the revenue figures before real use.
- Findings are associations, not proof of cause. A customer who calls care often may simply be a heavier user.
- Some data fields have heavy missing values, some column codes are undocumented, and the time period and contract terms are unknown.
- The profit example uses assumed margin, cost and success rates. Replace them with real figures.
- Ethnicity, marital status and income are sensitive. Use them for understanding only, never for pricing, offers or service decisions. Region and credit class can act as proxies, so review them before use.

Next steps

- Run Actions 1, 3 and 4 as a controlled test on a few thousand customers and compare with a control group.
- Measure real success rate, cost and margin, then rerun the profit calculation in Section 6.
- Score customers monthly and retrain the model as new data arrives.

Source: Telecom customer churn dataset (100,000 customers, 100 fields); analysis in Customer_Churn_Analysis.ipynb. Model performance metrics are reported in the notebook.